

Final Project Documentation

Airline Booking



1. Project Overview

This project focuses on analyzing an **Airline Booking Dataset** to uncover trends, customer preferences, timing patterns, geographic insights, and performance metrics.

The work was carried out in **three main environments**: Excel, Python (Pandas, Numpy, Matplotlib, Seaborn), and SQL Server.

The project follows a complete **data pipeline**:

1. Dataset sourcing and documentation.
2. Data cleaning and preprocessing.
3. Data transformation and feature creation.
4. Exploratory analysis and visualization.
5. SQL queries for answering business questions.
6. Final insights and reporting.

2. Dataset Documentation (Day 1)

Data Dictionary

Column Name	Description	Example Values	Data Type
num_passengers	Number of passengers in the booking	1, 2, 3	Integer
sales_channel	Booking channel	Online, Offline	String
trip_type	Type of trip	Round Trip, One Way, Circle Trip	String
purchase_lead	Days between booking date & travel date	5, 30, 60	Integer
length_of_stay	Days spent at destination	3, 7, 14	Integer
flight_hour	Hour of departure (0–23)	6, 14, 22	Integer
flight_day	Day of the week for departure	Monday, Friday	String
booking_origin	Country where booking was made	Egypt, USA	String

Column Name	Description	Example Values	Data Type
wants_extra_baggage	Extra baggage request	Yes, No	Boolean
wants_preferred_seat	Preferred seat request	Yes, No	Boolean
wants_in_flight_meals	In-flight meals request	Yes, No	Boolean
flight_duration	Flight duration in hours	2.5, 5.0	Float
booking_complete	Whether booking was completed	Yes, No	Boolean

3. Key Business Questions

The following questions guided the analysis:

Booking Trends

- Which sales channel has the highest number of bookings?
- How does booking completion rate vary between channels?

Customer Behavior

- What percentage of customers request extra baggage, preferred seats, or meals?
- What is the most common trip type?

Timing Analysis

- What is the average purchase lead time?
- Which days and hours are most popular for flight departures?

Geographic Analysis

- Which countries have the most bookings?
- What is the average length of stay by country?

Performance Metrics

- Average passengers per booking.
 - Average flight duration per trip type.
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4. Data Cleaning & Preprocessing (Day 2)

Excel Cleaning

- Checked for missing values → (no set → “Egypt”).
- Checked for duplicates → None found.
- Validated data types (numeric, categorical, boolean).
- Standardized Boolean values: Converted **0/1** to **Yes/No**.
- Standardized categories (consistent spelling).

Python Cleaning

- Used Pandas to re-check missing data and duplicates.
 - Confirmed no outliers in `purchase_lead`, `length_of_stay`, or `flight_duration`.
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5. Data Transformation

Added new calculated columns for analysis:

- **purchase_lead_group** → Categorized booking lead time into ranges (0–7 days, 8–30 days, More than 31+ days).
- **day_period** → Categorized `flight_hour` into Morning, Afternoon, Evening, Night.
- **day_type** → Classified `flight_day` as Weekend or Weekday.
- **booking** → Based on `booking_complete` status.

Changes to existing columns:

- `sales_channel`: Changed “Internet” & “Mobile” → **Online**, **Offline** retained.
 - Converted all service request columns (`wants_extra_baggage`, `wants_preferred_seat`, `wants_in_flight_meals`) from binary to Yes/No.
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6. Excel Analysis & Dashboard

Pivot Tables Created

- Booking Analysis by Sales Channel & Trip Type.
- Effect of Flight Time on Booking Completion
- Completed Booking Analysis.
- Chart Purchase Lead Time vs. Booking Completion.
- Flight Duration by Day & Period.

Charts Created

- Bar chart: Bookings by sales channel.
- Pie chart: Trip type distribution.
- Map chart: Bookings by country.
- Pie chart: Popular days & hours.

Dashboard

- Combined all pivot tables and charts into an interactive dashboard for quick insights.
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7. Python Analysis

Used Pandas, Numpy, Matplotlib, and Seaborn to answer key business questions:

- **Booking Trends** → Bar plots & count plots for sales channels and completion rates.
 - **Customer Behavior** → Pie charts for service requests & trip type.
 - **Timing Analysis** → Line plots for purchase lead distribution; heatmap for popular days/hours.
 - **Geographic Analysis** → Bar charts for top countries; boxplots for stay length.
 - **Performance Metrics** → Grouped bar charts for passengers per booking and duration per trip type.
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8. SQL Analysis

Dataset uploaded to **SQL Server**.

Queries were written to replicate Python/Excel analysis:

Example queries:

```
-- Booking Trends
-- 1.1
SELECT
    sales_channel,
    COUNT(*) AS booking_count
FROM [dbo].[Sheet1$]
GROUP BY [sales_channel]
ORDER BY booking_count DESC;
```

```
--4.2
SELECT
    booking_origin,
    AVG(flight_duration) AS avg_flight_duration
FROM [dbo].[Sheet1$]
GROUP BY booking_origin
ORDER BY avg_flight_duration DESC;
```

9. Project Pipeline

Step-by-Step Process:

1. Dataset sourcing & documentation (Day 1).
2. Data cleaning in Excel and Python.
3. Data transformation & feature engineering.
4. Exploratory data analysis in Excel (Pivot tables, charts, dashboard).
5. Python analysis & varied visualizations.
6. SQL queries for data validation & insights.
7. Final insights.

10. Key Insights (*Example*)

- Channel: Majority of bookings are (**Online** → 91.7%).
- Customer Preferences: **78.4%** request baggage, **48.5 %** meals, **30.9%** seats.
- Timing: Most flights depart on [**Monday**].
- Geography: Top origin countries: [**Australia, Malaysia, South Korea**].
- Performance: Avg. flight duration (**Medium** = **5.7 hrs**, **Long** = **8.42 hrs**, Avg. **passengers** = 1.4 ~ **1–2** passengers).

11. Tools & Skills Used

- **Excel**: Pivot Tables, Charts, Dashboards.
- **Python**: Pandas, Numpy, Matplotlib, Seaborn.
- **SQL Server**: Aggregations, GROUP BY, filtering.
- **Data Skills**: Cleaning, preprocessing, transformation, visualization, reporting.

12. Conclusion

The project demonstrates a **full data analysis pipeline**, starting from raw dataset collection to generating actionable business insights using multiple tools for cross-verification and efficiency.